Multivariate Analysis: Assessment 2

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Question 1

Table 1: Rank 4 Approximation Error

	Error
V1	0.0123508
V2	-0.0114648
V3	0.1320079
V4	0.0151170
V5	-0.0419993
V6	0.0124169
V7	0.0098078
V8	-0.0207314
V9	0.0605322
V10	-0.0433849
V11	0.1382238
V12	-0.0375508

Question 2

As see in Figure 1 *insert interpretation here*

Question 3

As see in Figure 2 the Frobenius norm decreases approximately linearly as k increases. This is expected as higher rank approximates contain more information about the dataset, x, and thus have a smaller error.

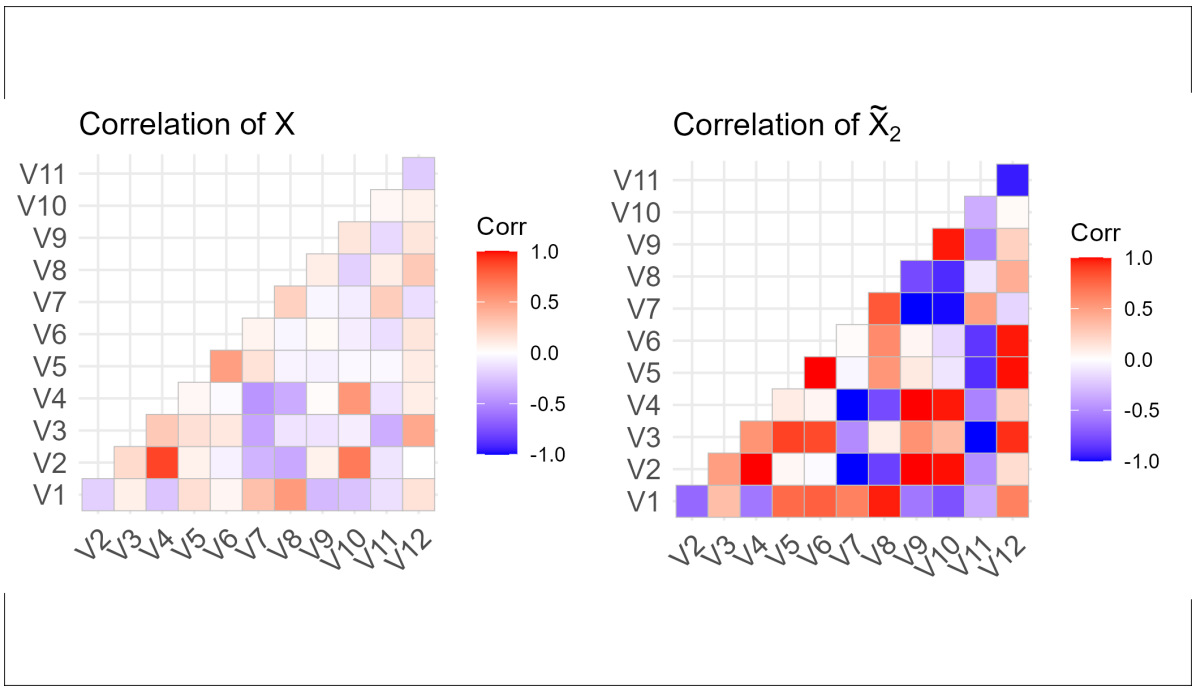

Figure 1: Correlation plots comparing X and the rank 2 approximation of X.

Question 4

As seen in Figure 3 the percentage of variation retained increases as k increases. This is expected as higher rank approximates contain more informartion about the original dataset and thus explain more variation in the data.

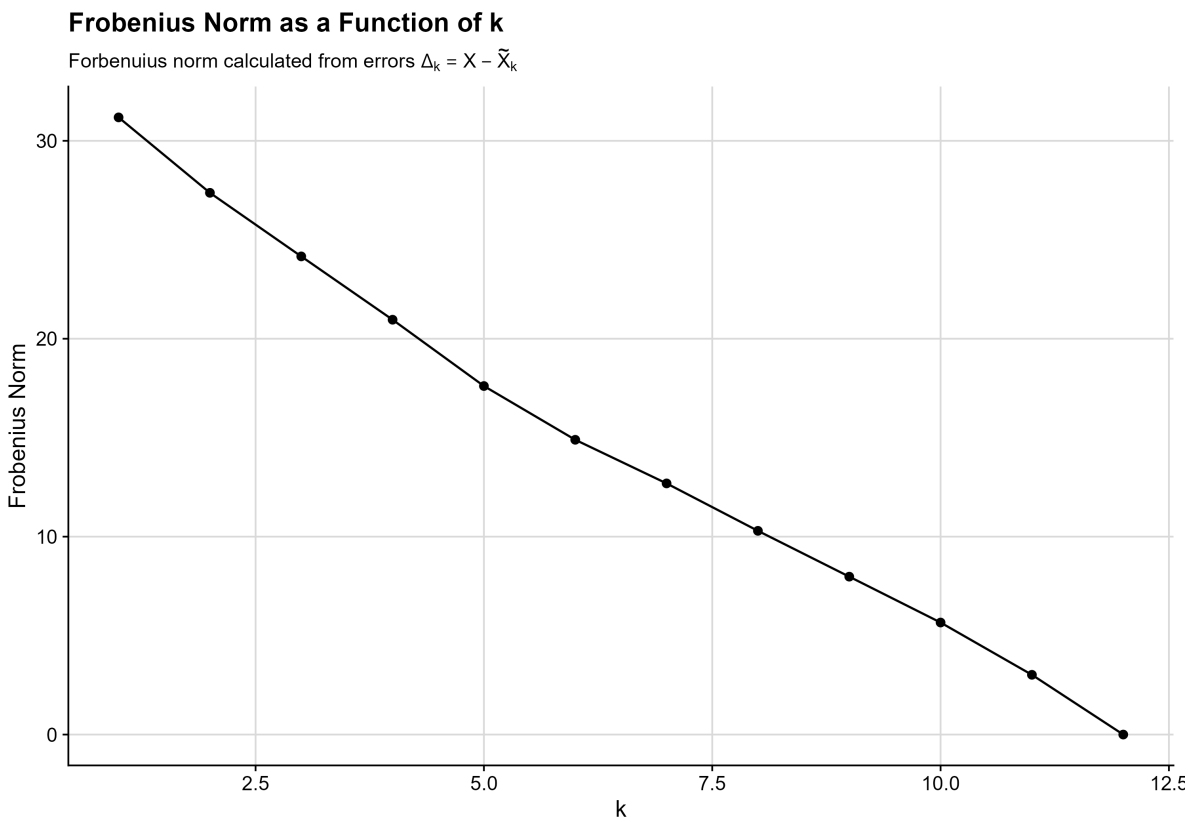

Figure 2: Frobenius Norm against k.

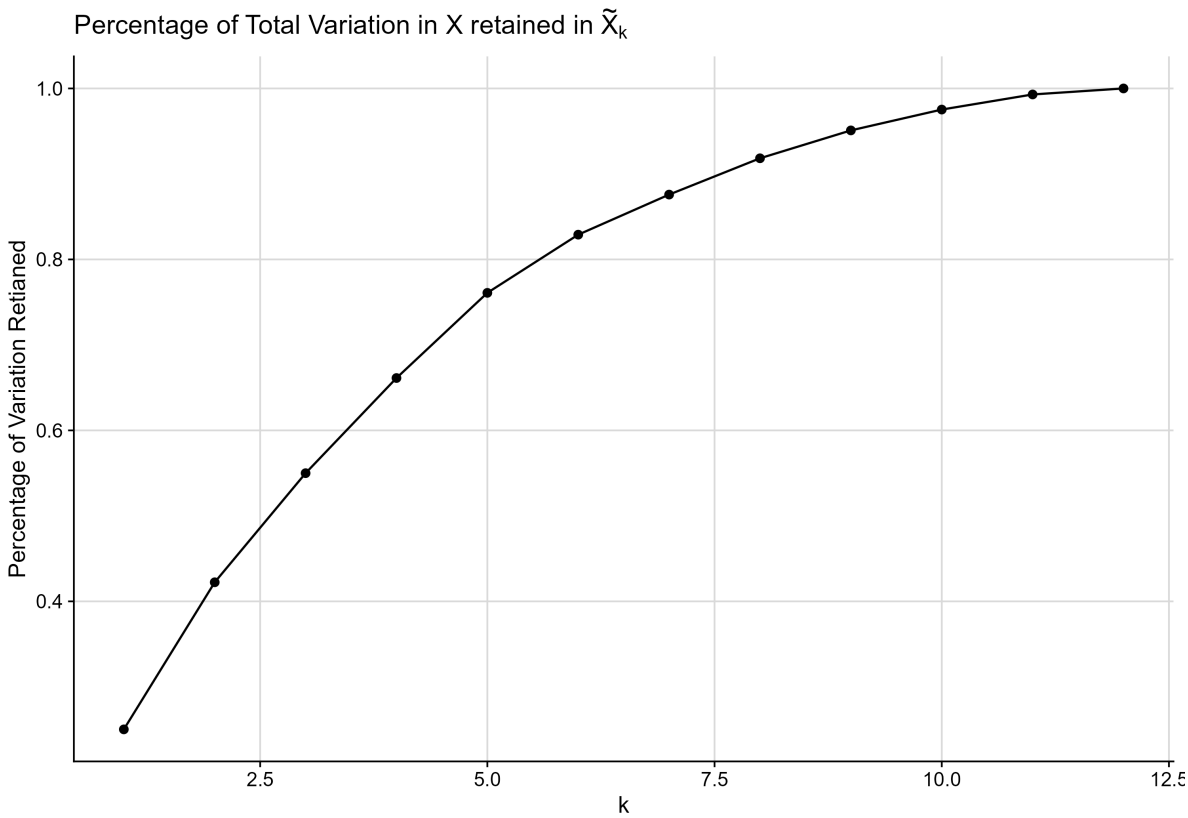

Figure 3: Percentage Variation Retained in X.